Teardown Benchmark Report FY23

Full-vehicle teardown and sub-system cost and mass benchmarking
Subject vehicle against two comparable-tonnage benchmarks

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Prepared by	VAVE Benchmarking Cell, Product Engineering
Reviewed by	Chief Engineer - Vehicle Integration
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Sub-systems covered	11
Components catalogued	2,418
Distribution	VAVE Committee; Product Engineering; Cost Engineering

Summary

A full-vehicle teardown was carried out on the subject vehicle and two comparable-tonnage benchmarks, cataloguing 2,418 components across eleven sub-systems. Eighty-four value-engineering opportunities were identified with a combined indicative benefit of INR 38,400 per vehicle. The engine ancillary group is reported in section 4 and carries the highest ratio of benefit to validation effort of any group in the study.

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1 Scope and method

This report covers a full-vehicle teardown of the subject vehicle against two comparable-tonnage benchmarks, carried out between October 2022 and January 2023. Every removable component was catalogued, weighed, dimensionally recorded and costed on a common basis. Eleven sub-systems were assessed; this revision reports the chassis, frame and engine ancillary groups in detail and summarises the remainder.

1.1 Costing basis

Piece costs are fully absorbed at an assumed annual volume of 16,000 units, expressed in Indian rupees at January 2023 input rates. Tooling is amortised over four years. Benchmark costs are engineering estimates derived from measured mass and observed process route; they carry an estimating tolerance of approximately 12 per cent and are not procurement intelligence.

1.2 How opportunities are classified

- Running change - no re-homologation, no new hard tooling, deliverable within the current model year.
- Platform change - requires validation, new tooling or a homologation activity, deliverable at the next platform refresh.
- Study - technically attractive but not costed or validated to the standard required for a committee submission.

Method note. An opportunity is counted once, against the sub-system where the change is made. Where a change benefits two sub-systems, the secondary benefit is noted in the text but not added to the total.

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2 Vehicles and teardown protocol

Attribute	Subject vehicle	Benchmark B	Benchmark C
Configuration	6x4 rigid	6x4 rigid	6x4 rigid
GVW (t)	28.0	28.0	28.0
Engine displacement (l)	5.9	6.4	5.1
Rated power (kW)	162	169	160
Odometer at teardown (km)	4,210	3,880	5,140
Components catalogued	812	798	808
Teardown duration (days)	9	9	8

2.1 Protocol

Each vehicle was drained, de-trimmed and stripped to bare frame over eight to nine working days. Components were cleaned, individually weighed on a 0.01 kg resolution scale, photographed and dimensionally recorded. Fasteners were counted and classified by grade and coating. Assemblies that could not be separated without destruction were recorded as assemblies and are marked as such in the catalogue.

2.2 Limitations of the method

- A teardown measures what was built, not why. Where a benchmark appears to carry unnecessary content, a load case not visible at teardown may justify it.
- Single-vehicle samples cannot separate design intent from build variation. Mass differences below 0.3 kg at component level are not reported as findings.
- Supplier process routes are inferred from part geometry and witness marks and are not confirmed with the suppliers.

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3 Chassis and frame

3.1 Frame side member

All three vehicles use a bolted ladder frame with a C-section side member. The subject vehicle's side member is 7 mm web against 6.5 mm on Benchmark B and 6 mm with a reinforcement insert on Benchmark C. The subject frame is 41.6 kg heavier than Benchmark C across the pair.

Component	Subject (kg)	Benchmark B (kg)	Benchmark C (kg)
Side member, pair	318.4	296.2	276.8
Crossmembers, total	84.6	79.1	81.2
Gussets and brackets	46.2	41.8	38.4
Frame fasteners	18.9	17.2	16.6
Frame group total	468.1	434.3	413.0

Benchmark C achieves its lighter frame by using a thinner base section with a local reinforcement insert over the bogie, rather than a uniform heavier section along the full length. The approach concentrates material where the bending moment is highest and is the same principle applied later in this report to the engine mounting brackets.

Finding 3.1. The frame side member carries an estimated 26 kg of material not required outside the bogie region. A reinforced thinner section is a platform change requiring frame re-homologation and is recorded as a study.

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3.2 Crossmembers and gussets

The subject vehicle uses six crossmembers against five on both benchmarks. The additional member sits between the gearbox and the first bogie hanger and was introduced in an earlier model year to address a frame resonance. Both benchmarks address the same frequency range with a tuned gusset at the hanger rather than an additional member.

Position	Subject	Benchmark B	Benchmark C	Comment
Front, behind cab	Channel, 6 mm	Channel, 6 mm	Channel, 5 mm	-
Gearbox support	Channel, 6 mm	Tubular	Tubular	Mass difference
Mid-span	Channel, 6 mm	Not fitted	Not fitted	Additional member
Bogie front	Channel, 8 mm	Channel, 8 mm	Channel, 6 mm + gusset	-
Bogie rear	Channel, 8 mm	Channel, 8 mm	Channel, 8 mm	-
Rear end	Channel, 5 mm	Channel, 5 mm	Channel, 5 mm	-

Finding 3.2. Deleting the mid-span crossmember and adopting a tuned hanger gusset would save 14.2 kg and INR 1,860 per vehicle. The frame resonance that justified the member must be re-measured before this can proceed, so it is classified as a platform change.

3.3 Summary of the chassis group

The chassis and frame group accounts for 41.6 kg and INR 4,120 of gap against the lightest benchmark. Neither of the two findings above is deliverable as a running change, which is characteristic of frame content: the validation burden is high and the lead time long.

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4 Engine ancillary sub-systems

The engine ancillary group covers the starter motor and its mounting, the alternator and its mounting, the air compressor drive, the power steering pump and the associated brackets, pulleys and belt tensioning hardware. It is reported in detail because it carries the highest ratio of benefit to validation effort of any group in this study: most of its content is bracketry that can be changed without affecting a homologated function.

Sub-group	Subject (kg)	Best benchmark (kg)	Gap (kg)	Opportunities
Starter motor and mounting	18.42	15.10	3.32	4
Alternator sub-system	14.86	11.24	3.62	3
Air compressor drive	11.20	10.86	0.34	1
Power steering pump and mount	8.64	7.92	0.72	2
Belt drive and tensioners	5.18	4.90	0.28	1
Group total	58.30	50.02	8.28	11

4.0.1 Why bracketry dominates the gap

In every sub-group above, the rotating machine itself is within 0.6 kg across the three vehicles; these are bought-in units of similar rating from a small supplier population. The gap is almost entirely in the brackets that carry them, which are designed in-house and are not subject to a supplier catalogue constraint.

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4.1 Starter motor and mounting

4.1.1 Machine

All three vehicles use a 24 V reduction-gear starter of similar rating. Measured masses are 12.84 kg on the subject vehicle, 12.60 kg on Benchmark B and 12.42 kg on Benchmark C. The 0.42 kg spread is within the range expected from supplier variation and is not a finding.

4.1.2 Mounting bracket

The subject vehicle mounts the starter on a cast-iron bracket of 5.58 kg. Benchmark B uses a cast-iron bracket of 4.21 kg with a ribbed rather than solid web. Benchmark C uses a fabricated steel bracket of 2.68 kg.

Attribute	Subject	Benchmark B	Benchmark C
Material	Grey cast iron	Grey cast iron	Fabricated steel
Construction	Solid web	Ribbed web	Pressed and welded
Mass (kg)	5.58	4.21	2.68
Machined faces	3	3	2
Estimated cost (INR)	1,840	1,610	1,180
Fixings	4 x M12	4 x M12	3 x M12

Finding 4.1. The starter mounting bracket carries 2.90 kg and INR 660 of gap against the lightest benchmark. A ribbed casting captures roughly half of it as a running change; a fabricated bracket captures all of it but requires a new supplier route.

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4.1.3 Load path and why the solid web persists

The solid web dates from an earlier engine installation in which the starter bracket also carried a harness support and a lifting eye. Neither is now fitted, and the bracket drawing has not been revisited since those features were deleted. This is a common pattern and is worth stating plainly: a component that was correctly specified for an earlier configuration may be over-specified for the current one, and nothing in the change process prompts a review.

> Finding 4.2. At least four components in this group retain features serving a deleted function. A periodic review of components whose surrounding installation has changed would surface these without a teardown.

4.1.4 Fixing arrangement

Benchmark C uses three M12 fixings against four on the subject vehicle, with the third relocated to react the torque reaction more directly. Measured bolt loads at crank are comparable. The saving is small in mass but removes an assembly operation and a part number.

4.1.5 Opportunities in this sub-group

Ref	Opportunity	Mass (kg)	Cost (INR)	Class
FY23-011	Ribbed web casting	1.37	230	Running change
FY23-012	Fabricated steel bracket	2.90	660	Platform change
FY23-013	Three-fixing arrangement	0.08	46	Running change
FY23-014	Delete redundant harness boss	0.21	38	Running change

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4.1.6 Pulley and drive interface

The starter drive interface is common across all three vehicles and presents no opportunity. It is recorded here for completeness because a proposal to change the ring gear tooth count was raised during the study and closed: the interface is defined by the engine supplier and is not within the platform's control.

Closed during the study. Ring gear tooth count is defined by the engine supplier specification and is outside platform control. Proposals to change it should be closed by reference to this paragraph.

4.1.7 Harness and connection

The subject vehicle uses a 70 sq mm starter cable against 50 sq mm on both benchmarks, over a comparable route length. The larger section dates from the same earlier installation as the bracket, when the battery was mounted further rearward. The current route is 0.9 m shorter.

Attribute	Subject	Benchmark B	Benchmark C
Cable section (sq mm)	70	50	50
Route length (m)	2.6	2.7	2.4
Cable mass (kg)	1.82	1.34	1.19
Measured volt drop at crank (V)	0.41	0.52	0.49

Finding 4.3. A 50 sq mm cable would save 0.48 kg and INR 310. The measured volt drop would rise from 0.41 V to an estimated 0.54 V, which remains within the cold-crank requirement. Classified as a running change subject to a cold-climate confirmation test.

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4.1.8 Heat shielding

The subject vehicle fits an aluminised steel heat shield of 0.84 kg between the starter and the exhaust downpipe. Benchmark B fits a smaller shield of 0.42 kg; Benchmark C fits none, relying instead on a 40 mm greater clearance achieved by routing the downpipe outboard.

Measured starter body temperature after a sustained grade climb was 118 degrees Celsius on the subject vehicle, 124 on Benchmark B and 109 on Benchmark C. The routing solution is therefore more effective than either shield, but it is a packaging change rather than a component change and cannot be adopted without a downpipe redesign.

Finding 4.4. Shield mass and routing are alternatives, not additive. A proposal to delete the shield without the routing change would raise starter body temperature by an estimated 19 degrees Celsius and should be rejected.

4.1.9 Summary, starter sub-group

Four opportunities totalling 4.56 kg and INR 974, of which three are running changes totalling 1.66 kg and INR 314. The sub-group illustrates the general pattern of this teardown: the machine is fixed, the bracketry is not, and the largest single contributor is a feature retained from a deleted function.

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4.1.10 Comparative cost build-up

Element	Subject (INR)	Benchmark B (INR)	Benchmark C (INR)
Starter motor	8,420	8,380	8,260
Mounting bracket	1,840	1,610	1,180
Fixings and hardware	186	184	140
Cable and terminals	1,240	930	880
Heat shield	310	190	0
Assembly labour	148	148	118
Sub-group total	12,144	11,442	10,578

The INR 1,566 gap against Benchmark C is 68 per cent bracketry, cable and shield, and 10 per cent the machine itself. This ratio is consistent across every sub-group in section 4 and is the reason the engine ancillary group was selected for detailed reporting.

4.1.11 Cross-reference

The bracket findings in this section are closely related to the alternator bracket discussed in section 4.2. Both are cast-iron parts carrying a bought-in machine, both retain features from earlier installations, and both are candidates for the same material and process change. They are reported separately because their load cases and thermal environments differ.

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4.2 Alternator sub-system

4.2.1 Mounting bracket - principal finding

The alternator is mounted on a cast-iron bracket of 4.86 kg on the subject vehicle. Benchmark C uses an HSLA steel fabricated bracket of 3.98 kg at equivalent stiffness. Replacing the cast-iron alternator mounting bracket with an HSLA steel bracket gives an estimated 18 per cent weight reduction with no change in fatigue life, on the basis of the comparative stiffness and the measured benchmark section.

Attribute	Subject	Benchmark B	Benchmark C
Material	Grey cast iron	Grey cast iron	HSLA steel, fabricated
Mass (kg)	4.86	4.52	3.98
Reduction against subject	-	7.0%	18.1%
First bending mode (Hz)	412	406	418
Estimated cost (INR)	1,460	1,390	1,120
Machined faces	2	2	1

Finding 4.5. Replace the cast-iron alternator mounting bracket with an HSLA steel bracket: approximately 18 per cent weight reduction with no change in fatigue life. The benchmark's measured first bending mode is higher than the subject vehicle's, so stiffness is not traded away.

4.2.2 Belt tensioning and pulley

The subject vehicle uses a manual tensioner with a slotted adjuster; both benchmarks use an automatic tensioner. The automatic unit is 0.34 kg heavier but removes a service adjustment and the associated warranty exposure from under-tensioning. It is not a mass or cost opportunity and is recorded as a quality observation.

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5 Cost and mass summary

Sub-system	Mass gap (kg)	Cost gap (INR)	Opportunities	Running changes
Chassis and frame	41.6	4,120	6	0
Engine ancillaries	8.3	4,860	11	7
Cab structure	18.4	6,240	14	5
Cab trim and interior	11.2	5,180	16	11
Suspension	22.8	4,920	9	2
Driveline	14.6	3,840	8	3
Brake system	6.4	2,680	7	4
Electrical and harness	4.2	2,410	6	4
Exhaust and aftertreatment	9.8	2,180	4	2
Fuel system	3.6	1,240	2	1
Cooling	2.9	740	1	1
Total	143.8	38,410	84	40

Finding 5.1. Forty of the eighty-four opportunities are running changes, totalling INR 14,820 and 52.4 kg per vehicle. The remaining forty-four require validation, tooling or homologation and are deliverable only at a platform refresh.

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6 Conclusions

- The total gap against the best comparable benchmark is 143.8 kg and INR 38,410 per vehicle across eleven sub-systems.
- Bracketry, not bought-in machines, accounts for most of the engine ancillary gap. The machines are within 0.6 kg across all three vehicles.
- A recurring cause is features retained from deleted functions: at least four components in the ancillary group carry content for an earlier installation.
- The alternator bracket finding in section 4.2 is the clearest single opportunity in the group at 18 per cent mass reduction with no fatigue consequence.
- Frame content dominates the mass gap but delivers none of it as a running change, because the validation burden is high and the lead time long.

Recommendation. Release the forty running changes to the VAVE committee as a single batch, and open a review of components whose surrounding installation has changed since their last drawing revision.

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